Empirical Calibration for Interval-Level Variances of MDLQ Methane Emission Rate Estimates

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1 Introduction

The multisource detection, localization, and quantification (MDLQ) model estimates methane emission rates using point-in-space continuous monitoring observations and an atmospheric transport model within a Bayesian framework [1]. There are known sources of error that are not accounted for in the MDLQ, causing the posterior variance of the emission rate parameters and the resulting nominal coverage to be too low. For example, the MDLQ uses the Gaussian puff atmospheric dispersion model to forward transport methane from source locations to sensor locations, but this model does not account for turbulence or blockage. In fact, a simulation study in earlier work demonstrated that artificial (but still realistic) forward model errors reduced credible interval coverage to levels seen in a real data evaluation [1]. In the absence of a higher fidelity forward model, and to account for other unmodeled sources of error, we provide an empirical adjustment to the variance of the emission rate parameters from the original MDLQ. Empirical uncertainty calibration has precedent in the methane emissions literature. For example, certain aerial technologies use controlled release data to characterize and correct uncertainties in their emission rate estimates [2]. Similar efforts to empirically assess and calibrate uncertainties also appear in satellite retrieval research [3].

We recalibrate interval-level variances using controlled releases at the Methane Emissions Technology Evaluation Center (METEC). Specifically, we fit a single parameter to scale the posterior standard deviations of the interval-level emission rate parameters, with its value selected to minimize disagreement between empirical and nominal interval coverage.

2 Data and Methods

We use the same data as the original MDLQ paper: MDLQ output and controlled release records from the 2024 Advancing Development of Emissions Detection (ADED) campaign that took place at METEC. This campaign comprised 337 controlled releases conducted between February 5 and April 30, 2024, with ten point-in-space methane sensors monitoring five equipment groups. Equipment-level release rates ranged from 0.08 to 7.2 kg h^{-1}, and experiments included emissions from multiple

sources simultaneously [1]. By default, the MDLQ segments the entire experiment into 30-minute nonoverlapping windows and provides a separate emission rate estimate for each window. True release rates are averaged over the corresponding windows, including periods with no emissions.

Our calibration procedure is as follows. We define information periods for a given source as periods when wind conditions transport potential emissions from that source toward the sensor network, allowing observations to inform its emission state and rate. During no-information periods, potential emissions are transported away from the sensors, so the absence of a measured enhancement cannot establish that the source is not emitting. For each interval i, we retain sources with positive, nonmissing MDLQ emission rate estimates, excluding sources without estimates due to a lack of information or other causes. Denoting this set of sources by $\mathcal{S}_i$, and letting $\hat{q}$ and q^{true} represent the interval-level MDLQ emission rate point estimate (posterior mean) and the true release rate, respectively, the estimated and true totals across $\mathcal{S}_i$ are given by:

$$\widehat{Q}_i = \sum_{j \in \mathcal{S}_i} \hat{q}_{ij}, \qquad Q_i^{\text{true}} = \sum_{j \in \mathcal{S}_i} q_{ij}^{\text{true}}.$$

We apply the same source selection to the posterior samples, and for posterior draw m calculate

$$Q_i^{(m)} = \sum_{j \in \mathcal{S}_i} q_{ij}^{(m)}, \qquad s_i^2 = \mathbb{V}\text{ar}_m \left(Q_i^{(m)} \right).$$

Summing within each posterior draw is done because source-level estimates tend to be biased, whereas estimates aggregated across sources are unbiased [1]. Intervals without eligible sources are excluded. Both estimated and true totals include only the selected sources, $\mathcal{S}_i$, so this procedure excludes emissions from sources assigned zero or missing estimates.

We construct normal uncertainty intervals centered on $\widehat{Q}_i$, with recalibrated variance $a^2 s_i^2$. At nominal coverage probability p, the lower and upper bounds are

$$L_i(p; a) = \max \left\{ 0, \widehat{Q}_i - z_{(1+p)/2} a s_i \right\}, \quad \text{and} \quad U_i(p; a) = \widehat{Q}_i + z_{(1+p)/2} a s_i \,,$$

respectively, where z_u is the standard normal quantile at probability u. Lower bounds are truncated at 0, as we do not allow for negative emission rates.

We can then calculate empirical coverage as the proportion of intervals containing the corresponding true total emission rate:

$$\widehat{C}(p; a) = \frac{1}{N} \sum_{i=1}^{N} \mathbb{1} \left[L_i(p; a) \leq Q_i^{\text{true}} \leq U_i(p; a) \right],$$

where N is the total number of intervals with at least one source in $\mathcal{S}_i$, and $\mathbb{1}$ is the indicator function.

We evaluate 37 nominal coverage probabilities from 0.05 to 0.95 in increments of 0.025. The multiplier is selected by a grid search over $a \in \{0, 0.025, ..., 6.975, 7\}$, minimizing the root mean squared error (RMSE) between empirical and nominal coverage:

$$\hat{a} = \arg\min_a \sqrt{\frac{1}{37} \sum_{k=1}^{37} \left[\hat{C}(p_k; a) - p_k \right]^2}.$$

3 Results

The selected multiplier was $a = 4.5$, corresponding to a 4.5-fold increase in posterior standard deviations, and a 20.25-fold increase in posterior variances. The 37 evaluated nominal coverage probabilities and their corresponding empirical coverages using recalibrated uncertainties, along with the pre-calibration uncertainties for comparison, are shown in Figure 1. The coverage curve approaches the 1-to-1 line at intermediate nominal probabilities, but retains systematic departures from nominal coverage: empirical coverage exceeds nominal coverage at lower probabilities but falls below it at higher probabilities. For example, nominal 95% uncertainty intervals achieve 83.3% empirical coverage after calibration. The selected multiplier minimizes coverage RMSE within the evaluated grid, but does not achieve nominal coverage at every probability level.

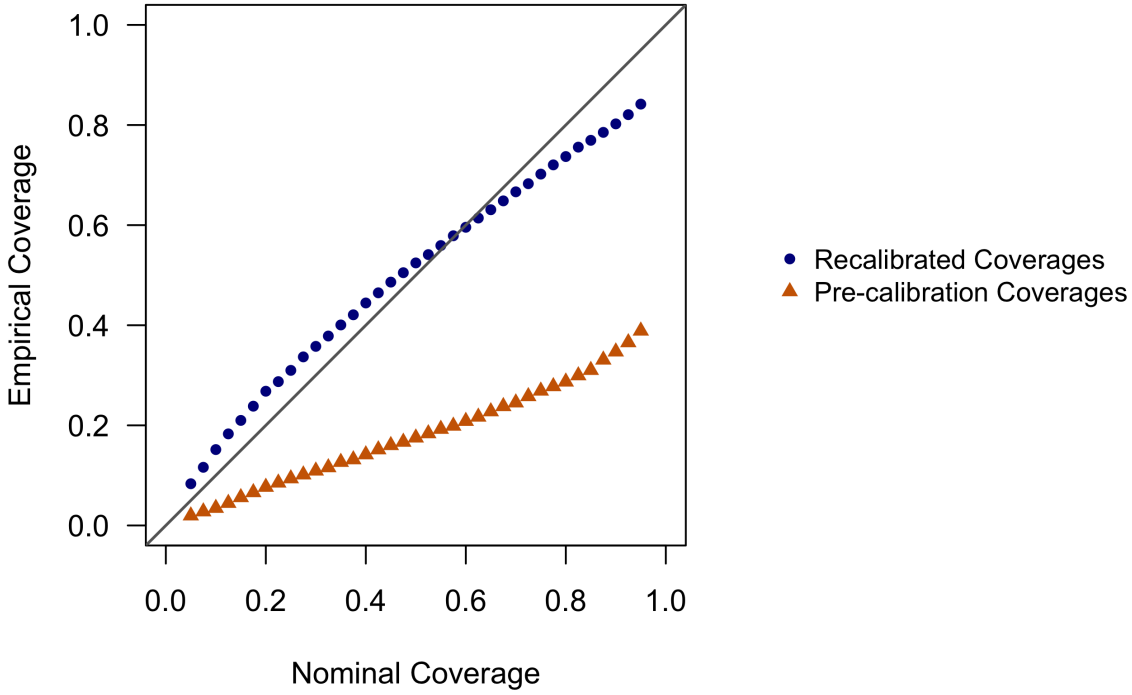


Figure 1: Empirical coverage versus nominal coverage for pre-calibration ($a = 1$) and recalibrated ($a = 4.5$) intervals. Points represent the 37 evaluated nominal coverage probabilities; the solid black line represents exact agreement. Blue circles represent recalibrated coverages, while orange triangles represent pre-calibration coverages.

Figure 2 shows a sample of the METEC controlled release rates, along with the MDLQ estimates and associated 95% uncertainty intervals. Credible intervals before recalibration are shown with solid vertical lines, while uncertainty intervals after recalibration are shown with transparent vertical lines. Few of the before-recalibration credible intervals contain the ground truth, emphasizing the need for this sort of recalibration in our application.

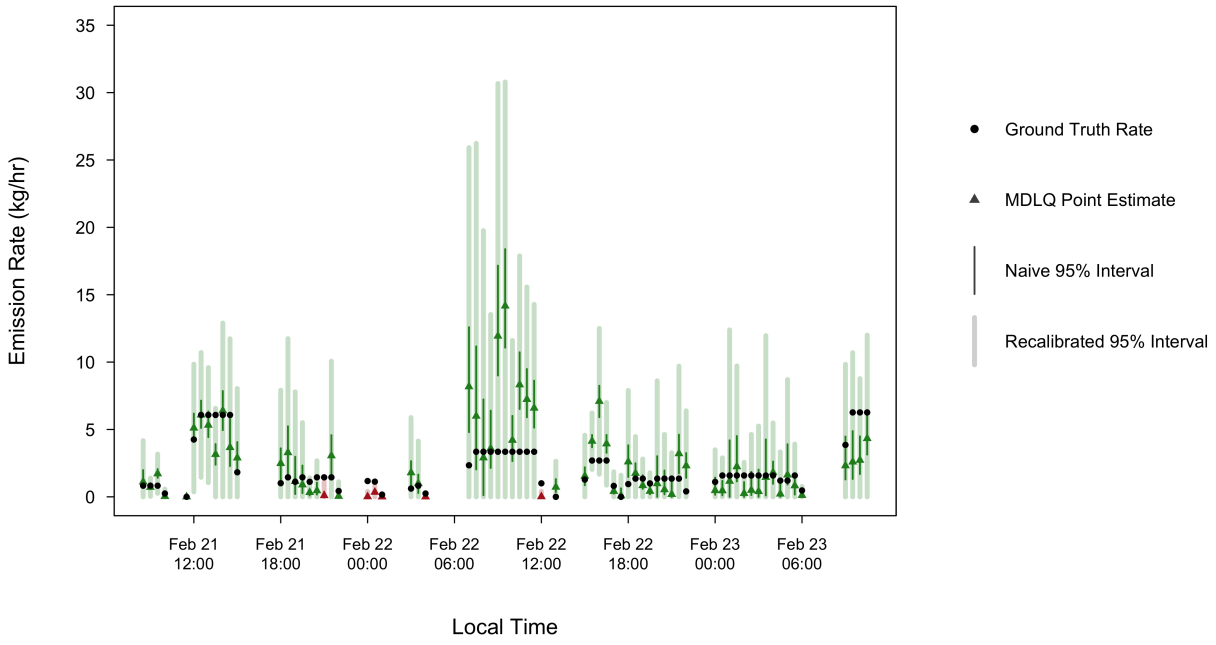


Figure 2: Time series showing ground truth rates (black dots), MDLQ point estimates (green/red triangles), and both naive and recalibrated 95% intervals. Color of MDLQ estimates indicate whether the recalibrated 95% uncertainty interval does (green) or does not (red) contain the ground truth.

Controlled-release data provide an empirical basis for adjusting MDLQ variances to account for unmodeled sources of error, especially forward model error. Scaling posterior standard deviations by 4.5 provides a practical uncertainty adjustment while retaining the original emission rate estimates. Although undercoverage remains at higher nominal probabilities, the selected multiplier minimizes overall coverage RMSE on the calibration data. Future work will involve dropping the normal assumption and directly rescaling the posterior distribution while preserving its shape and retaining the asymmetric nature of the pre-calibration credible intervals.

References

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